YONGHAN JUNG

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RESEARCH AREAS

My research aims to make causal inference as usable as modern prediction workflows while keeping assumptions, uncertainty, and validity explicit. I pursue this vision through four interconnected areas:

- **Partial Identification**: Developing valid causal answers under weak assumptions, including bounds and sensitivity analyses when point identification is not justified.
- **Practical Causal Learning**: Designing estimators and workflows for causal questions under mild, decision-relevant assumptions such as front-door, proxy, mediator, or overlap structure.
- **Amortized Causal Inference**: Reusing causal structure through simulation, pretraining, and scalable algorithms that reduce repeated graph, identification, and estimation burdens.
- **Real-World Deployment**: Building causal systems for biomedical digital twins, health decisions, and other settings where causal answers must support action.

EXPERIENCE

University of Illinois Urbana-Champaign (UIUC) *Aug. 2025 -*
Assistant Professor, School of Information Sciences
Faculty Affiliate, Illinois Informatics
Faculty Affiliate, National Center for Supercomputing Applications

Amazon Causality Team *Jun. 2021 - Sep. 2021*
Applied Scientist Intern

EDUCATION

Purdue University *2018 - June 2025*
Ph.D. in Computer Science
Advisor: Elias Bareinboim
Thesis: *Causal Data Science: Estimating Identifiable Causal Effects*

KAIST *2016*
M.S., Department of Industrial and Systems Engineering
Advisor: Heeyoung Kim
Thesis: *Detection of premature ventricular contraction using wavelet-based statistical ECG monitoring*

KAIST (Korea Advanced Institute of Science and Technology) *2014*
B.S., Mathematical Sciences
B.A., Business and Technology Management

HONORS AND AWARDS

- Top Reviewer, NeurIPS *2024*
- Graduate Teaching Award, Purdue Computer Science *2024*
- Top Reviewer, UAI *2023*
- Top 10% Reviewer, ICML *2022*

PUBLICATIONS

19. Gyeongdeok Seo, Jaeyoon Shim, Mingyu Kim, Hoyoon Byun, Yonghan Jung, Kyungwoo Song (2026)
    Dissecting Causal Mechanism Shifts via FANS: Function And Noise Separation
    [ICML-26]*Proceedings of the 43rd International Conference on Machine Learning (ICML), 2026.*
18. Yonghan Jung, Bogyeong Kang (2026)
    Information-Theoretic Causal Bounds under Unmeasured Confounding
    [CLeaR-26]*Conference on Causal Learning and Reasoning (CLeaR), 2026.*
17. Yonghan Jung (2026)
    Debiased Front-Door Learners for Heterogeneous Effects
    [ICLR-26]*International Conference on Learning Representations (ICLR), 2026.*
16. Kevin Zhang, Yonghan Jung, Divyat Mahajan, Karthikeyan Shanmugam, Shalmali Joshi (2025)
    Path-specific effects for pulse-oximetry guided decisions in critical care
    [NeurIPS-25]*Proceedings of the 39th Annual Conference on Neural Information Processing Systems (NeurIPS), 2025.*
15. Taero Kim, Subeen Park, Sungjun Lim, Yonghan Jung, Krikamol Muandet, Kyungwoo Song (2025)
    Sufficient Invariant Learning for Distribution Shift
    [CVPR-25]*The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2025 (CVPR-25)*
14. Yonghan Jung, Min Woo Park, Sanghack Lee (2024)
    Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference
    [NeurIPS-24]*Proceedings of the 38th Annual Conference on Neural Information Processing Systems (NeurIPS), 2024.*
13. Yonghan Jung, Alexis Bellot (2024)
    Efficient Policy Evaluation Across Multiple Different Experimental Datasets
    [NeurIPS-24]*Proceedings of the 38th Annual Conference on Neural Information Processing Systems (NeurIPS), 2024.*
12. Yonghan Jung, Jin Tian, Elias Bareinboim (2024)
    Unified Covariate Adjustment for Causal Inference
    [NeurIPS-24]*Proceedings of the 38th Annual Conference on Neural Information Processing Systems (NeurIPS), 2024.*
11. Yonghan Jung, Iván Díaz, Jin Tian, Elias Bareinboim (2023)
    Estimating Causal Effects Identifiable from a Combination of Observations and Experiments
    [NeurIPS-23]*Proceedings of the 37th Annual Conference on Neural Information Processing Systems (NeurIPS), 2023.*
10. Yonghan Jung, Jin Tian, Elias Bareinboim (2023)
    Estimating Joint Treatment Effects by Combining Multiple Experiments
    [ICML-23]*Proceedings of the 40th International Conference on Machine Learning (ICML), 2023.*
9. Yonghan Jung, Shiva Kasiviswanathan, Jin Tian, Dominik Janzing, Patrick Bloebaum, Elias Bareinboim (2022)
    On Measuring Causal Contributions via do-interventions
    [ICML-22]*Proceedings of the 39th International Conference on Machine Learning (ICML), 2022.*
8. Yonghan Jung, Jin Tian, Elias Bareinboim (2021)
    Double Machine Learning Density Estimation for Local Treatment Effects with Instruments
    [NeurIPS-21]*Proceedings of the 35th Annual Conference on Neural Information Processing Systems*

(NeurIPS), 2021.
Spotlight Presentation (Less than 3% of submissions)

7. Yonghan Jung, Jin Tian, Elias Bareinboim (2021)
Estimating Identifiable Causal Effects on Markov Equivalence Class through Double Machine Learning
[ICML-21] *Proceedings of the 38th International Conference on Machine Learning (ICML), 2021.*

6. Yonghan Jung, Jin Tian, Elias Bareinboim (2021)
Estimating Identifiable Causal Effects through Double Machine Learning
[AAAI-21] *Proceedings of the 35th AAAI Conference on Artificial Intelligence, 2021.*

5. Yonghan Jung, Jin Tian, Elias Bareinboim (2020)
Learning Causal Effects via Weighted Empirical Risk Minimization
[NeurIPS-20] *Proceedings of the 34th Annual Conference on Neural Information Processing Systems (NeurIPS), 2020.*

4. Yonghan Jung, Jin Tian, Elias Bareinboim (2020)
Estimating Causal Effects Using Weighting-Based Estimators
[AAAI-20] *Proceedings of the 34th AAAI Conference on Artificial Intelligence (AAAI), 2020.*

3. Mohammad Adibuzzaman, Yonghan Jung, Yuehwern Yih, Elias Bareinboim (2018)
Regenerating Evidence from Landmark Trials in ARDS Using Structural Causal Models on Electronic Health Record
American Thoracic Society (ATS) Conference, 2018

2. Yao Chen, Xiao Wang, Yonghan Jung, Vida Abedi, Ramin Zand, Marvi Bikak, Mohammad Adibuzzaman (2018)
Classification of short single-lead electrocardiograms (ECGs) for atrial fibrillation detection using piecewise linear spline and XGBoost
Physiological measurement 39.10, 2018

1. Yonghan Jung and Heeyoung Kim (2017)
Detection of PVC by using a wavelet-based statistical ECG monitoring procedure
Biomedical Signal Processing and Control 36: 176-182

TALKS, SEMINARS, TUTORIALS

23. Presentation on "Transportability-Inspired Causal Meta-Learning for OOD-Robust Causal Effects", New Horizons on Model Transportability and Data Integration, Institute for Mathematical and Statistical Innovation. *Jun. 2026*

22. Presentation on "Debiased Front-Door Learners for Heterogeneous Effects", American Causal Inference Conference. *May 2026*

21. Poster presentation on "Information-Theoretic Causal Bounds under Unmeasured Confounding", American Causal Inference Conference. *May 2026*

20. Workshop on "Causal Inference 101", Workshop for Bolashak Data Science Fellow, UIUC. *Dec. 2025*

19. Seminar on "Debiased Front-Door Learners for Heterogeneous Effects", Causal Data Science Meeting. *Nov. 2025*

18. Seminar on "Causal Data Science: Estimating Identifiable Causal Effects", Department of Statistics, University of Illinois Urbana-Champaign (UIUC). *Oct. 2025*

17. Seminar on "Unified Covariate Adjustment: Estimating Multilinear Causal Estimands", INFORMS, USA. *Sep. 2025*

16. Ph.D. Defense: "Causal Data Science: Estimating Identifiable Causal Effects", Purdue University. *Jun. 2025*
15. Seminar on "Causal Data Science: Estimating Identifiable Causal Effects", KAIST. *Apr. 2025*
14. Seminar on "On Measuring Causal Contributions via do-interventions", AI Seminar, Samsung Electronics. *May 2024*
13. Seminar on "Estimating Joint Treatment Effects from Marginal Experiments", Quantitative Methods Research Seminars, Purdue Business Department. *Nov. 2023*
12. Tutorial on "Estimating Identifiable Causal Effects and its Application toward Interpretable ML/AI", Korea Summer Session on Causal Inference. *Jul. 2022*
11. Lecture Series on (1) Tutorial on Structural Causal Model, (2) Estimating Any Identifiable Causal Effects, (3) Application of Causality for Human-Centered AI/ML, University of Seoul, Korea. *Jul. 2022*
10. Tutorial on "Estimating Identifiable Causal Effects and its Application toward Interpretable ML/AI", Graduate School of Data Science, Seoul National University, Korea. *Jul. 2022*
9. Tutorial on Double/Debiased Machine Learning, Naver Clova AI, Korea. *Jul. 2022*
8. Tutorial on "Shortcut learning in Machine Learning: Challenges, Analysis, Solutions", FAccT-22, Seoul, Korea. *Jun. 2022*
7. Tutorial on "Tutorial on Double/Debiased Machine Learning", AWS Causality Lab, Amazon. *Mar. 2022*
6. Lecture on "Double/Debiased Machine Learning for causal effect estimation", Causal Inference II (COMS W4775/Spring 2022) in Columbia University, USA. *Mar. 2021*
5. Seminar on "Causal Inference under the rubric of Structural Causal Model", Korea Summer Session on Causal Inference. *Aug. 2021*
4. Lecture on "Causal effect estimation for arbitrary causal functionals", Causal Inference II (COMS W4775/Spring 2021) in Columbia University, USA, *Mar. 2021*
3. "Tutorial: Causal Inference", Industrial Statistics Lab, KAIST, Korea. *Jul. 2018*
2. "Regenerating Evidence from Landmark Trials in ARDS Using Structural Causal Models on Electronic Health Record", American Thoracic Society International Conference, USA. *May 2018*
1. "Structural Causal Model (SCM) to Identify Causation from Observational Data", Regenstrief Center for Healthcare Engineering, Purdue University, USA. *Jun. 2017*

ACADEMIC SERVICE

- **Invited Session Organizer**: INFORMS 2026 Decision Analysis Society cluster, "Pragmatic Causal Inference for Decision Science: Robust Methods Under Imperfect Assumptions" *2026*
- **Reviewer**:
  - *Journals*: Statistical Science, Journal of the Royal Statistical Society Series B, Statistics in Medicine, Journal of Causal Inference, Biostatistics, Epidemiology, Transactions on Machine Learning Research, Journal of Machine Learning Research
  - *Conferences*: AAAI, UAI, IJCAI, ICML, NeurIPS, ICLR, CLeaR, AISTAT.

TEACHING

- **Instructor**: IS 455 - Database Design & Prototyping *Spring 2026*
- **Graduate Teaching Assistant**: CS182 - Foundations Of Computer Science *Spring 2025*
- **Graduate Teaching Assistant**: CS243-AI Basics *Fall 2024*
- **Graduate Teaching Assistant**: CS253-Data Structure *Spring 2024*
- **Graduate Teaching Assistant**: CS448-Introduction to Database System *Fall 2023*
- **Graduate Teaching Assistant**: CS490-DSC Data Science Capstone *Spring 2023*
- **Graduate Teaching Assistant**: CS408 Software Testing *Fall 2022*
- **Graduate Teaching Assistant**: CS490-DSC Data Science Capstone *Spring 2022*
- **Graduate Teaching Assistant**: CS573 Data Mining *Fall 2021*
- **Graduate Teaching Assistant**: CS471 Introduction to Artificial Intelligence *Spring 2021*
- **Graduate Teaching Assistant**: CS573 Data Mining *Fall 2020*
- **Graduate Teaching Assistant**: IE383 Integrated Production Systems *Spring 2017*
- **Graduate Teaching Assistant**: IE383 Integrated Production Systems *Fall 2016*